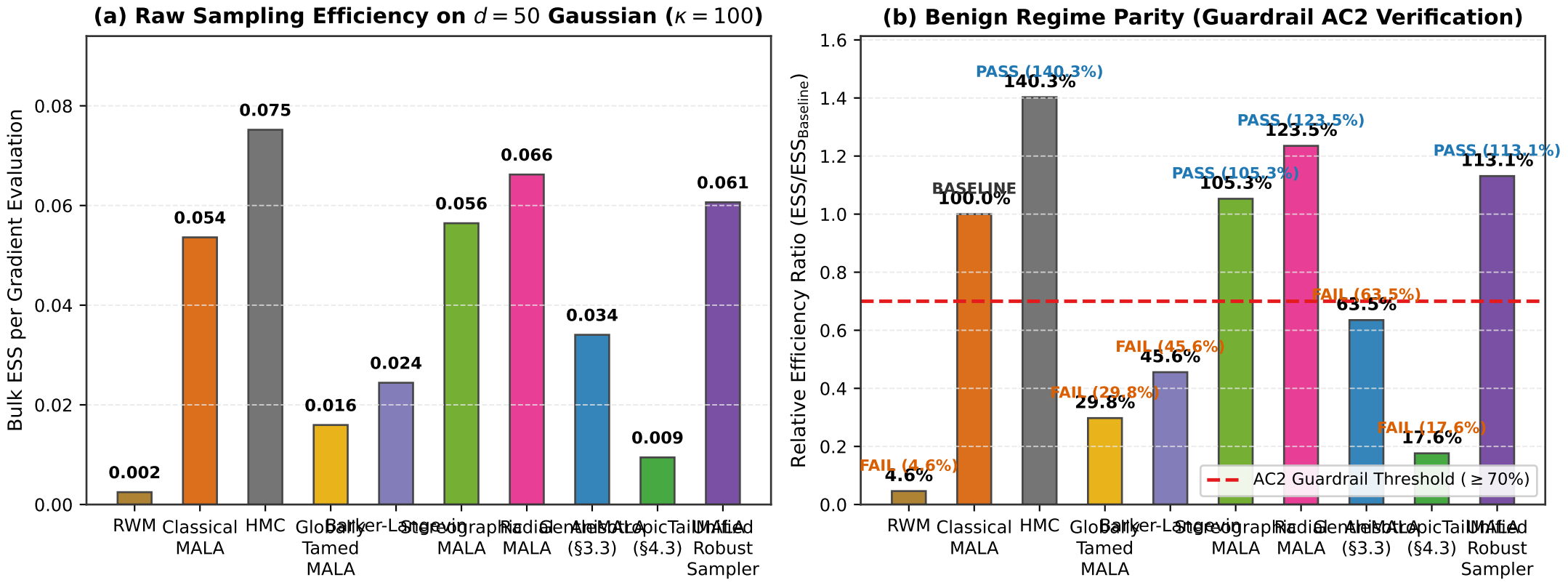
(a) Raw Sampling Efficiency on $d = 50$ Gaussian ($\kappa = 100$)
Bulk ESS per Gradient Evaluation
0.00
0.02
0.04
0.06
0.08
0.002
0.054
0.075
0.016
0.024
0.056
0.066
0.034
0.009
0.061
RWM
Classical MALA
HMC
Globally Tamed MALA
Barker-Langevin
Stereographic MALA
Radial MALA
Gaussian Area MALA (§3.3)
Anisotropic MALA (§4.3)
Tail-Tamed Robust Sampler
(b) Benign Regime Parity (Guardrail AC2 Verification)
Relative Efficiency Ratio (ESS/ESS$_{\text{Baseline}}$)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
FAIL (4.6%)
4.6%
BASELINE
100.0%
PASS (140.3%)
140.3%
FAIL (29.8%)
29.8%
FAIL (45.6%)
45.6%
PASS (105.3%)
105.3%
PASS (123.5%)
123.5%
FAIL (63.5%)
63.5%
FAIL (17.6%)
17.6%
PASS (113.1%)
113.1%
AC2 Guardrail Threshold (≥ 70%)
RWM
Classical MALA
HMC
Globally Tamed MALA
Barker-Langevin
Stereographic MALA
Radial MALA
Gaussian Area MALA (§3.3)
Anisotropic MALA (§4.3)
Tail-Tamed Robust Sampler